UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/01**

Paper 1 Multiple Choice **May/June 2009**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

* 2 7 3 7 0 4 9 3 1 5 *

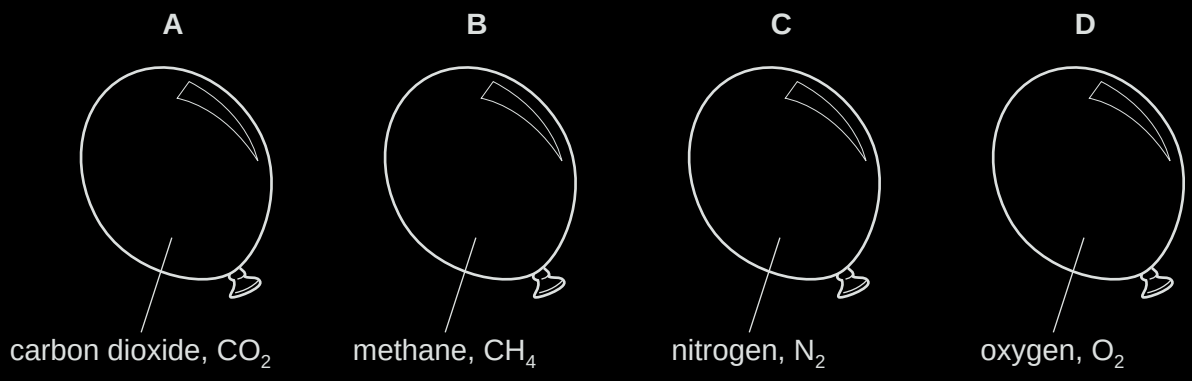
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

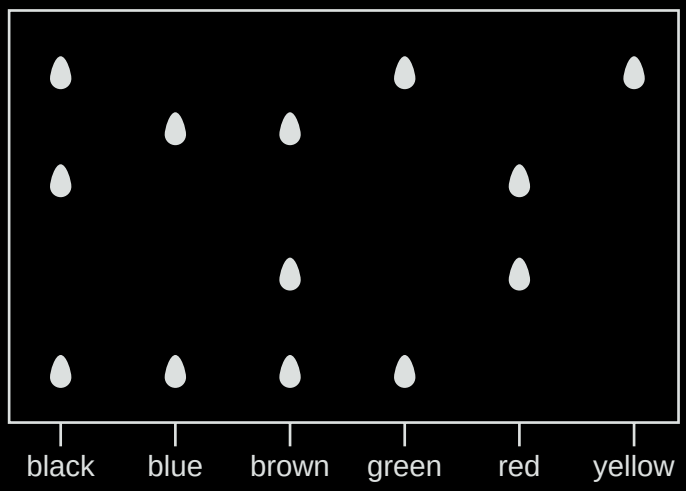
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
A copy of the Periodic Table is printed on page 16.



2 The diagram shows a chromatogram of several inks.



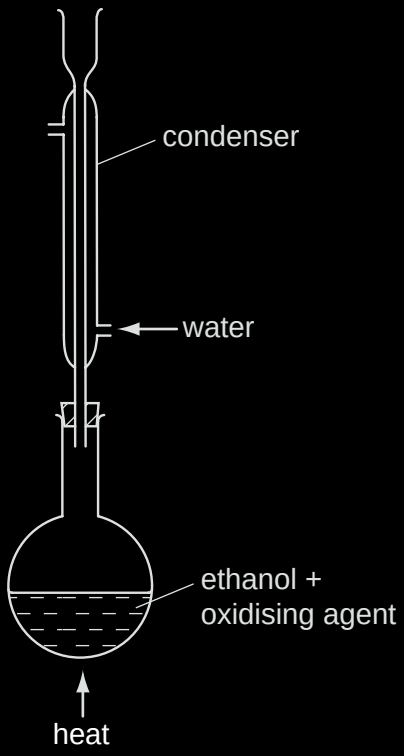
Which statement is correct?

A Black ink can be made by mixing green, red and yellow inks.

B Brown ink can be made by mixing blue and red inks.

C Yellow ink can be used to make brown ink.

D Yellow ink may be present in green ink.



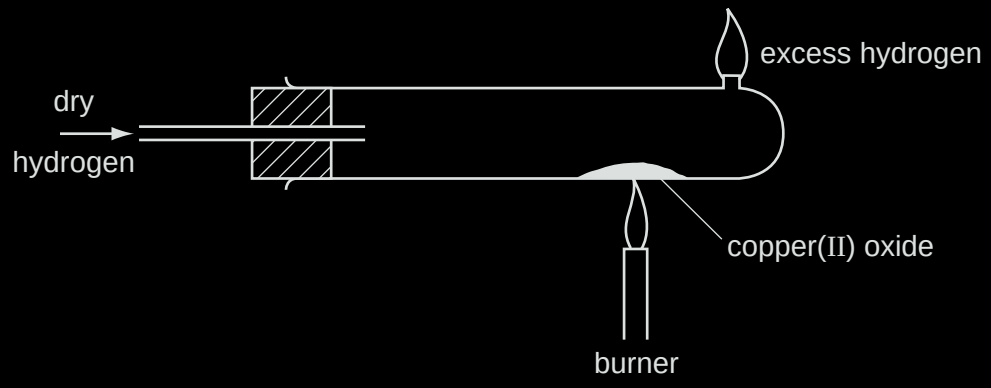
What is the purpose of the condenser?

A to prevent air reacting with the ethanoic acid

B to prevent any ethanol from escaping

C to prevent the ethanoic acid changing back to ethanol

D to prevent the ethanoic acid reacting with the ethanol



Why is the hydrogen allowed to flow through the tube during cooling?

A to allow the tube to cool slowly

B to lessen the risk of explosion in the hot tube

C to prevent the copper from reacting with the air

D to remove any traces of water left in the tube

5 A coin is analysed by dissolving it in nitric acid. To the resulting solution an excess of aqueous ammonia is added and the mixture is filtered.

A brown precipitate remains in the filter paper and a deep blue solution is obtained as the filtrate.

Which metals does the coin contain?

A aluminium and copper

B copper and iron

C iron and lead

D lead and zinc

6 An element X forms a positive ion with the electronic structure 2,8,8.

What is the proton (atomic) number of X?

A 16 **B** 17 **C** 18 **D** 19

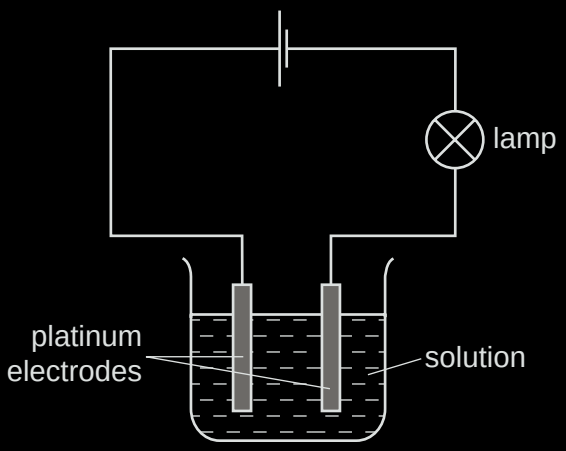
7 Which two substances are elements with a giant molecular structure?

A diamond and graphite

B diamond and sand

C methane and iodine

D methane and sand



Which substance, in aqueous solution of concentration 1 mol / dm^3, would cause the lamp to give the brightest light?

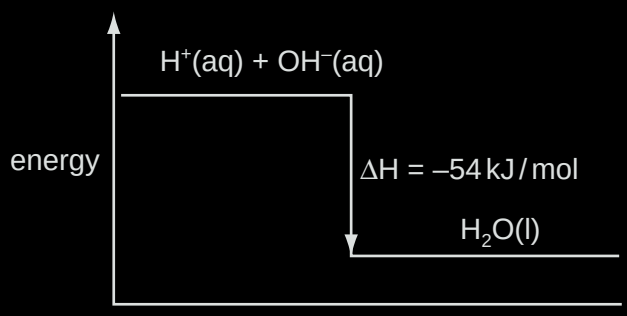
A ammonia

B ethanoic acid

C ethanol

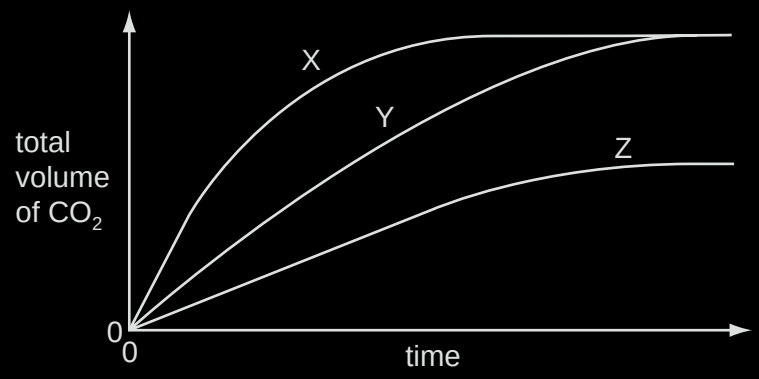
D sulfuric acid

15 The energy diagram for the reaction between sodium hydroxide and hydrochloric acid is shown.



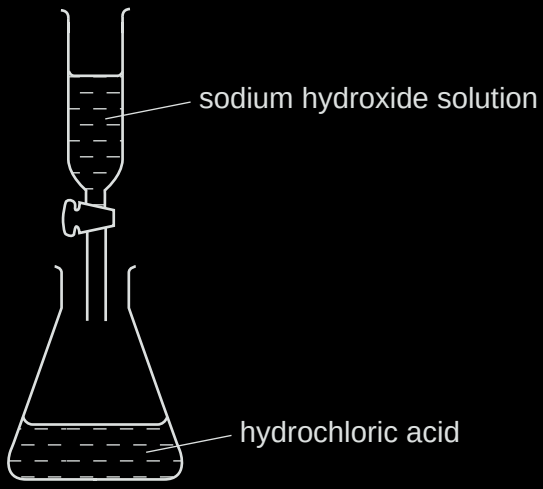
Which quantity of heat is liberated when 100 cm^3 of 1 mol / dm^3 hydrochloric acid reacts with 100 cm^3 of 1 mol / dm^3 sodium hydroxide?

A 0.54 kJ **B** 2.70 kJ **C** 5.40 kJ **D** 10.8 kJ



Which pair of curves is obtained in the two experiments?

	experiment 1	experiment 2
A	X	Z
B	X	Y
C	Y	Z
D	Y	X



What would be the pH change in this reaction?

A decrease to 1

B decrease to 7

C increase to 7

D increase to 12

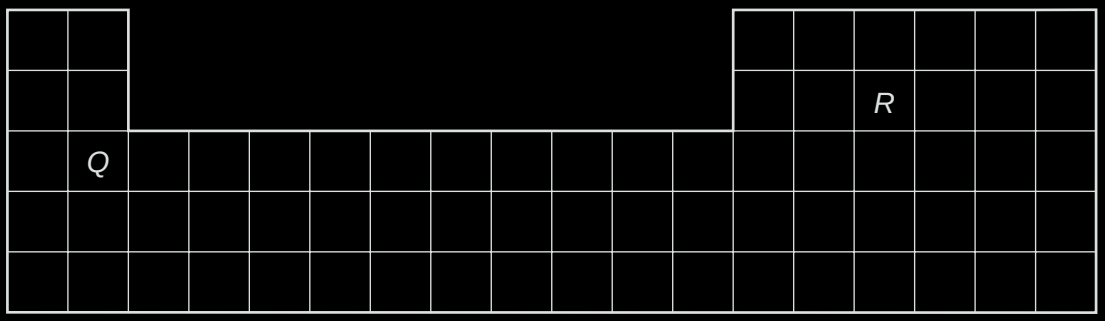
21 Why is nickel used in the addition of hydrogen to alkenes?

A It increases the yield of products.

B It lowers the activation energy of the reaction.

C It makes the reaction more exothermic.

D It prevents a reverse reaction from occurring.



What is the formula of the compound formed between them?

A QR_2 **B** Q_2R **C** Q_2R_3 **D** Q_3R_2

24 The list shows some properties of metals.

1 Metals are good conductors of electricity.

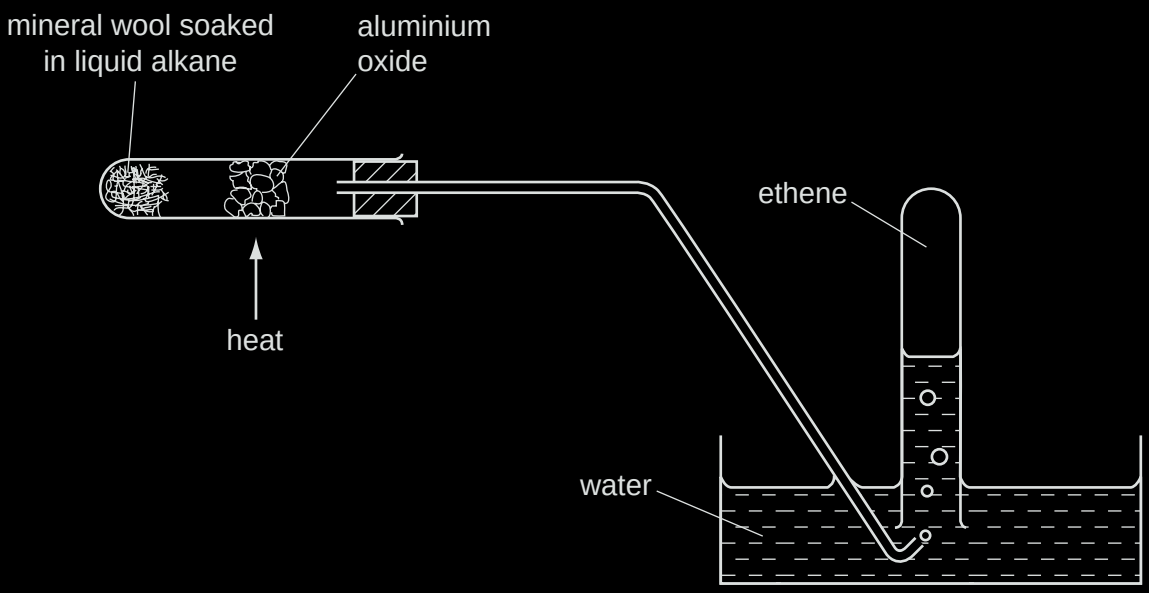
2 Metals form ions by the loss of electrons.

3 Metals have high melting points.

Mercury is a metallic element.

Which of these statements do **not** apply to mercury?

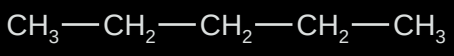
A 1 only **B** 1 and 2 **C** 2 and 3 **D** 3 only



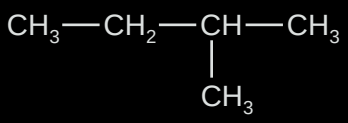
The ethene is then tested with aqueous bromine.

Which information about ethene is correct?

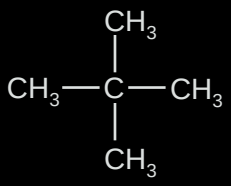
	solubility of ethene gas	action on aqueous bromine
A	insoluble	decolourised
B	insoluble	no reaction
C	soluble	decolourised
D	soluble	no reaction



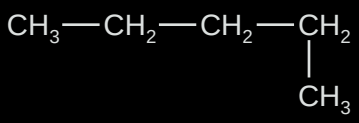
A



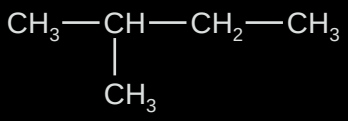
B



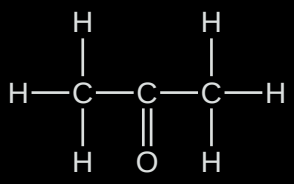
C



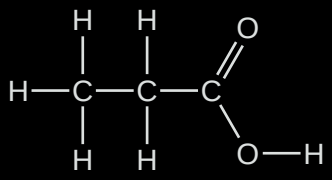
D



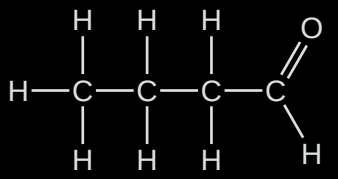
A



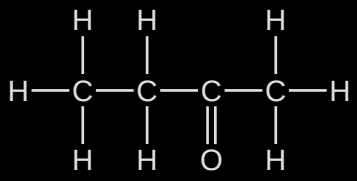
B



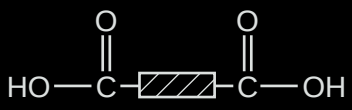
C



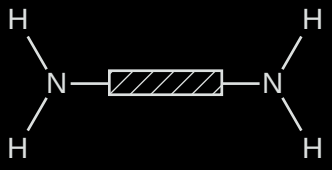
D



40 A polymer X is hydrolysed and the two products are



and



What can be deduced about X?

A It is a condensation polymer.

B It is made by addition polymerisation.

C It is starch.

D It is *Terylene*.

DATA SHEET
The Periodic Table of the Elements

Group																	
I	II										III	IV	V	VI	VII	0	
							1 H Hydrogen 1									4 He Helium 2	
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18	
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).